

COMPSCI5079 (M)

Cryptography & Secure Development

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Lecture 5:

Public-key Cryptography



Lecture 4:

Review





Main points to take home

➤ Message Digests

- Use a one-way hash function.
- Resistant to a birthday attack (collision)
- Open competition for SHA-3

➤ Random Numbers Generators

- Real random numbers based on keyboard latency time or physical fluctuations
- Pseudo-random numbers are predictable but have random statistics.
- Cryptographically secure pseudo-random numbers - knowledge of part of the sequence can't generate the rest.



Main points to take home

- Secret Sharing / Multiple-key cryptography.
 - Secret splitting using XOR. All parties needed to reconstruct the key.
 - Random bit strings, R , S and T , and the $P = M \oplus R \oplus S \oplus T$
 - The original message is recovered by $P \oplus R \oplus S \oplus T = M$.
 - Secret sharing based on linear equations. Only some of the parties needed
 - Shamir Secret Sharing Scheme (Threshold scheme)
 - If 2 shadows are needed to construct the key, then the appropriate polynomial is a line which has two unknown coefficients a and b :

$$y(x) = a * x + b \pmod{p}$$

 - » p : large prime number, made public
 - » Random number coefficient $a < p$
 - » The coefficient b is the secret message M .

Lecture 5:

Public-key Cryptography





Outline

1. Overview

- Introduction
- A Big Picture

2. Diffie Hellman Key Exchange Protocol

3. RSA Public key Cryptography

- KeyGen
- Encryption / Decryption
- Digital Signature

4. Elliptic Curves Cryptography

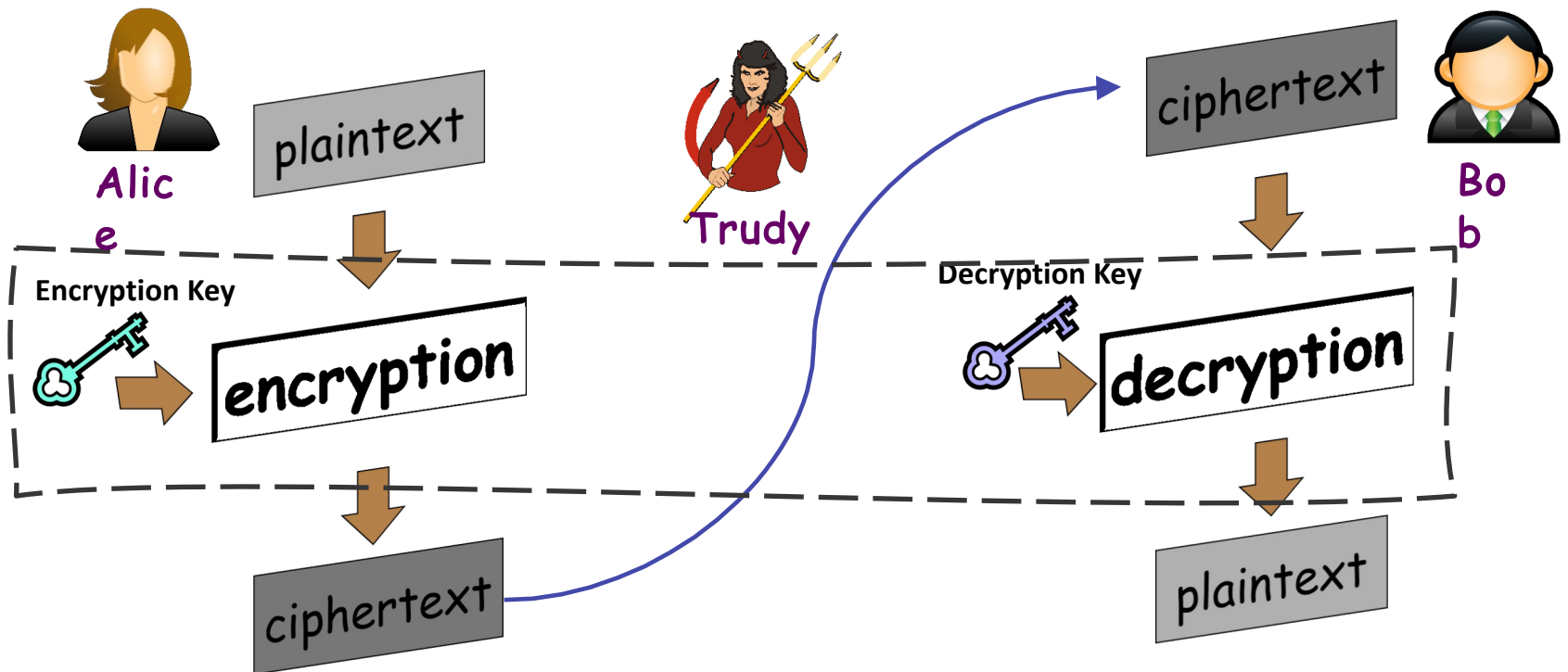
Lecture 5:

1. Overview





How Cryptography work





What we have learned so far

- Single-key / Symmetric key Systems:
 - Alice and Bob both know a same secret K so that the Encryption key and Decryption key can be inferred from the secret key K
 - For instance: Substitution Cipher:

Plaintext: abcdefghijklmnopqrstuvwxyz

Ciphertext: mnbvcxzasdfghjklpoiuytrewq

- The secret key, in this case, is the "substitution pattern"

Plaintext: bob. i love you. alice

Ciphertext: nkn. s gktc wky. mgsbc



Public Key Cryptography

Symmetric Key Crypto

Key Distribution challenge:
Requires sender, receiver
know shared secret key:

Q: How to agree on the key
in the first place
(particularly if never "met")?

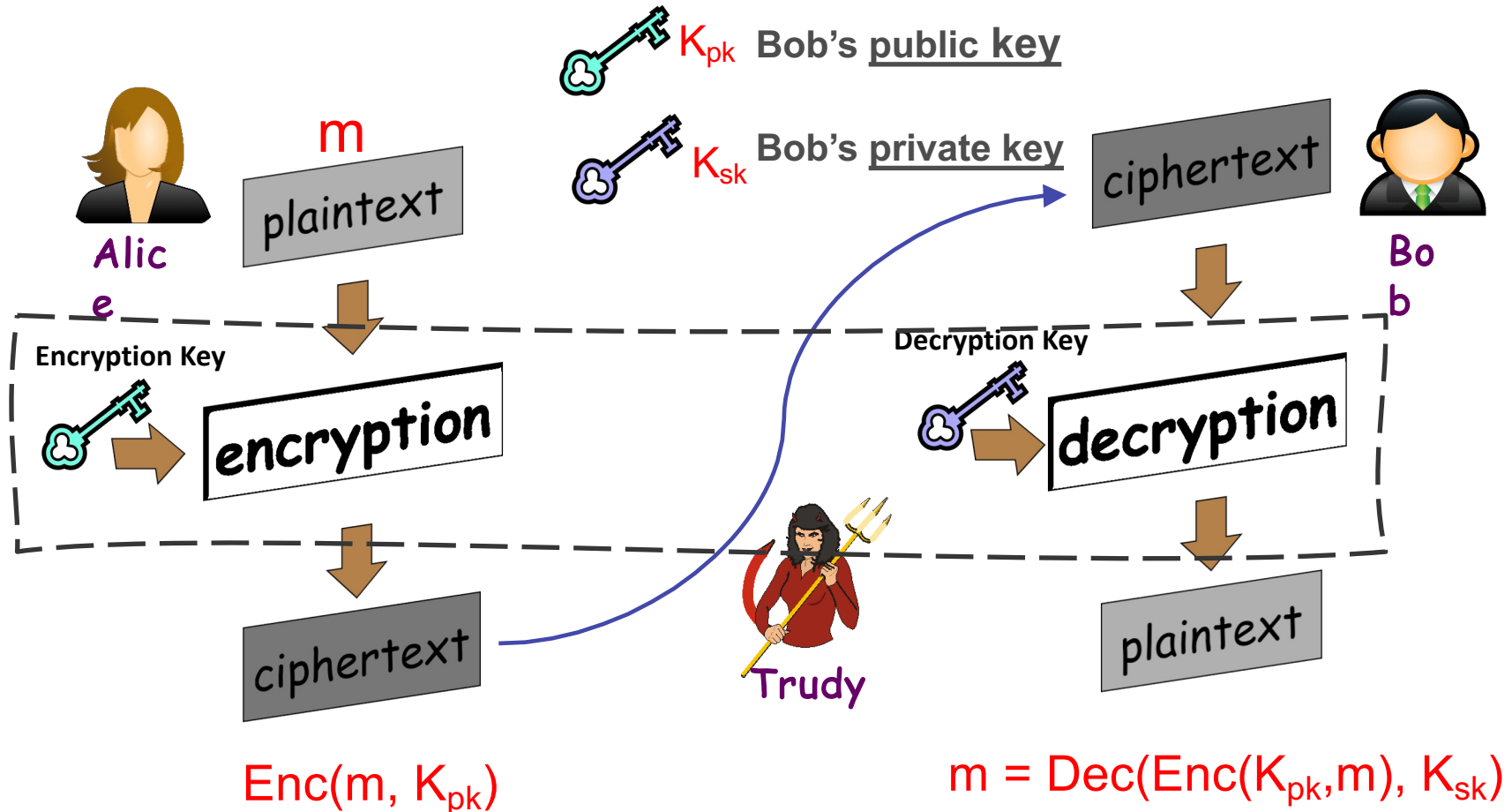
Public Key Cryptography

- Radically different approach [Diffie-Hellman76, RSA78]
- Sender, receiver do *not* share secret key
- *Public* encryption key known to *all*
- *Private* decryption key known only to receiver





How Cryptography works





Public-key Cryptography: Requirements

- Each person owns a key-pair (K_{sk} , K_{pk}).
 K_{pk} revealed to the world; K_{sk} is kept secretly
- Given Public key K_{pk} , it is impractical to compute the private key K_{sk}
- Need to design functions **Enc** and **Dec** so that:

$$m = \text{Dec} (K_{sk}, \text{Enc} (K_{pk}, m))$$

- Note that: "Use public key first (encrypt), followed by private key (decrypt)" AND "Use private key first, followed by public key" should produce the same results.



Public-key Cryptography: Overview

- Foundation of today's secure communication
- Allows communicating parties to obtain a shared secret key
- Public key (for encryption) and Private key (for decryption)
- Private key (for digital signature) and Public key (to verify signature)



Brief History

- Historically same key was used for encryption and decryption
 - Challenge: key distribution, how to exchange key
- 1976: Whitfield Diffie and Martin Hellman
 - Key exchange protocol
 - This indeed proposed a new public-key cryptosystem
- 1978: Ron Rivest, Adi Shamir, and Leonard Adleman
 - Attempted to develop a cryptosystem
 - Created RSA algorithm

Lecture 5:

2. Diffie-Hellman Key Exchange Protocol





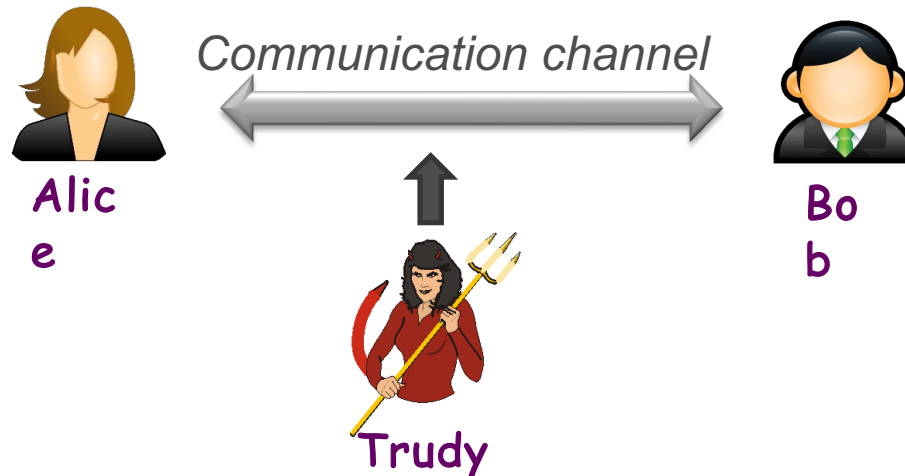
Key Exchange Challenge

- One of the main problems of symmetric key encryption is it requires a secure & reliable channel for the shared key exchange.
- The Diffie-Hellman Key Exchange protocol offers a way in which a public channel can be used to create a confidential shared key
- **Diffie-Hellman Key Exchange:** a protocol that enables two users to establish a secret key using a public-key scheme based on discrete logarithms.
 - The protocol is secure only if the authenticity of the two participants can be established.
- In practice, the shared encryption key relies on such complex concepts as *Modular Exponentiation*, *Primitive Roots* and *Discrete Logarithm Problems*



Diffie Hellman idea

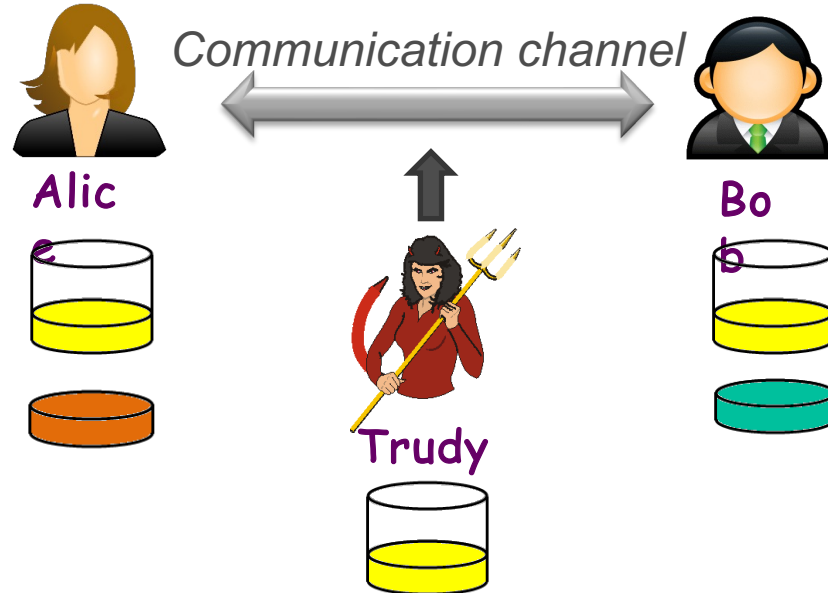
- A Difficult One-Way Operation: easy to DO but hard to UNDO
 - Mixing Colors: Easy to mix 2 colours, hard to unmix
 - Purpose: Alice & Bob (with Trudy listening) wish to make a **secret shared colour**





Diffie Hellman idea

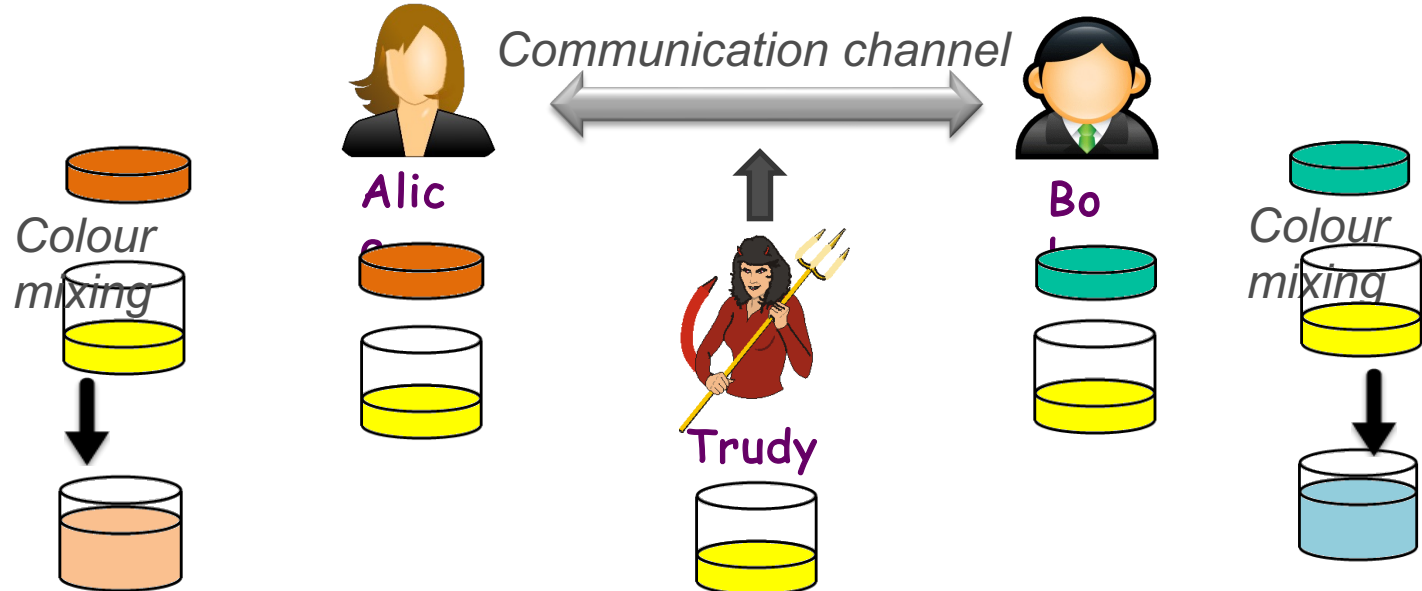
- Step 1: Both publicly agree to a shared colour
 - Let's say it's YELLOW, and Trudy can obtain this colour by listening to the communication channel





Diffie Hellman idea

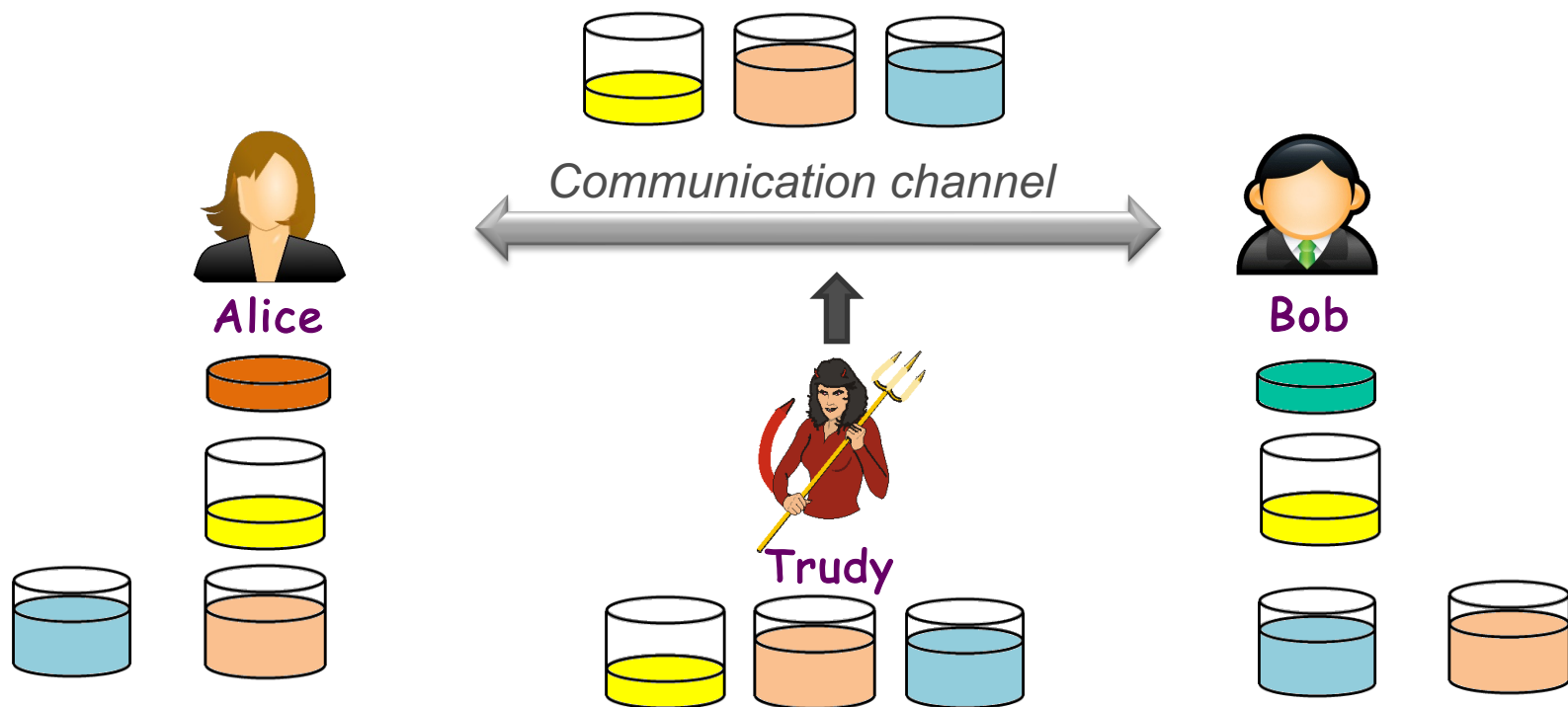
- Step 2 - Each picks a secret colour
 - Alice: ORANGE, Bob: TEAL
- Step 3 - Each adds (mixes) their secret colour to the shared colour





Diffie Hellman idea

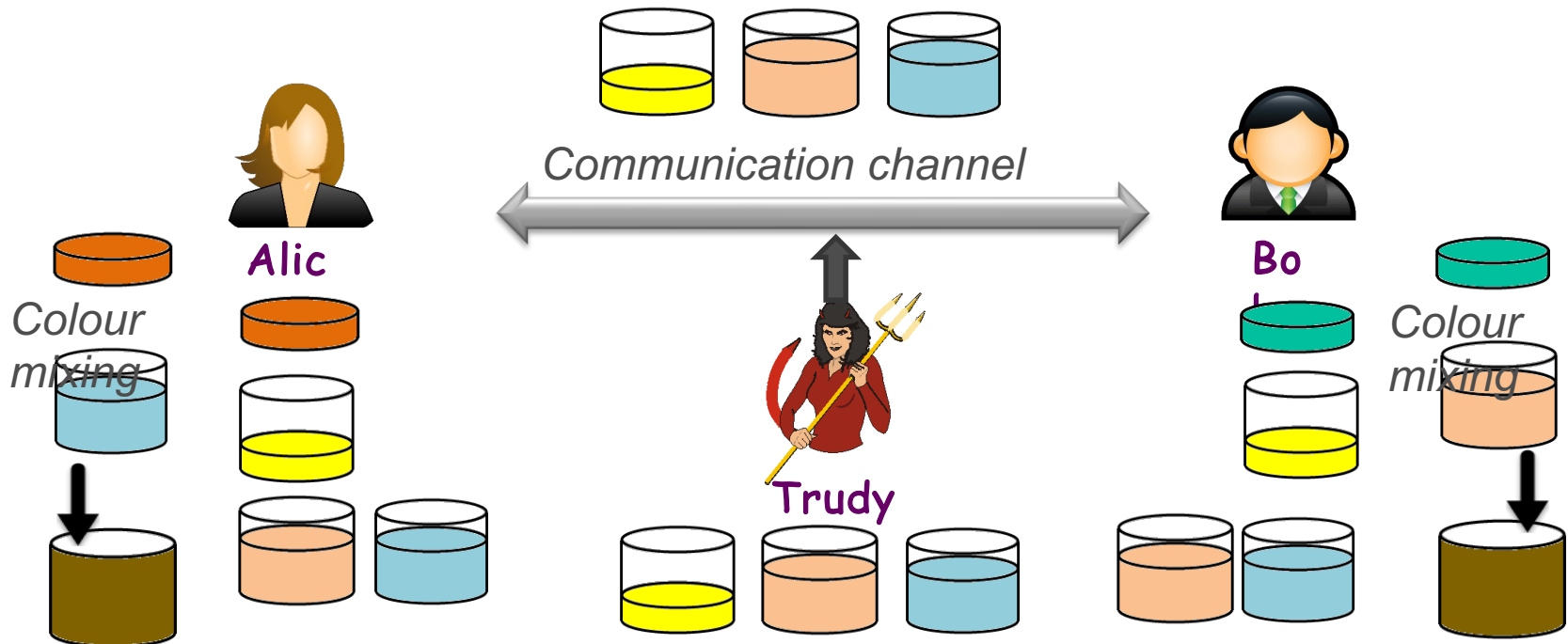
- Step 4 - Each sends the other their new mixed color





Diffie Hellman idea

- Step 5 - Each combines the shared colour from the other with their own secret colour
- Result: Alice and Bob successfully established a secret shared colour





Alice & Bob have agreed to a shared colour unknown to Trudy

➤ How is it that Alice & Bob's final mixtures are identical?

✓ Alice mixed:

✓ $[(\text{Yellow} + \text{Teal})_{\text{from Bob}}] + \text{Orange}$

✓ Bob mixed:

✓ $[(\text{Yellow} + \text{Orange})_{\text{from Alice}}] + \text{Teal}$



Alice & Bob have agreed to a shared colour unknown to Trudy

- How is it that Alice & Bob's final mixture is secret?
 - ✓ Trudy never has knowledge of the secret colours of either Alice or Bob
 - ✓ Unmixing a colour into its component colours is a hard problem



Diffie Hellman Algorithm: Overview

- The Diffie-Hellman key exchange uses an exponential encryption system to generate a single key that is shared by two people.
- Both parties contribute to the key by sharing information over an insecure communication channel.
- Each party keeps some information secret. This secret information is vital to be able to construct the key.
- The key cannot be generated from the public information.
- This algorithm forms the basis for session key construction in many internet applications.



One-way Operation

- We need to find an operation similar to "Mixing colours": Easy to DO and hard to UNDO.

Let's consider Multiplicative group of Integers mod p

$$s = g^n \bmod p$$

- We have:
Easy: given g , n , & p , solve for s
Hard: given s , g , & p , solve for n
- And the property of:

$$g^{a*b} \bmod p = g^{b*a} \bmod p$$



Step 1—Publicly shared information

- Alice & Bob publicly agree to a large prime number called the modulus, or p .
- Alice & Bob publicly agree to a number called the generator, or g , which has a primitive root relationship with p .
- In our example we'll assume $p = 17, g = 3$
- Trudy is aware of the values of p or g .



Step 2—Select a Secret Key

- Alice selects a secret key, which we will call **a**.
- Bob selects a secret key, which we will call **b**.
- For the example, we assume:
 $a = 54$
 $b = 24$
- Trudy is unaware of the values of **a** or **b**.



Step 3-Combine secret keys with public information

- Alice combines her secret key of a with the public info to compute A as her public key.

$$A = g^a \bmod p$$

$$A = 3^{54} \bmod 17$$

$$A = 15$$

- Bob combines his secret key of b with the public info to compute B as his public key.

$$B = g^b \bmod p$$

$$B = 3^{54} \bmod 17$$

$$B = 16$$



Step 4–Share combined values

- Alice shares her combined value, A , with Bob. Bob shares his combined value, B , with Alice.
- Sent to Bob
 $A = 15$
- Sent to Alice
 $B = 16$
- Eve is privy to this exchange and knows the values of A and B



Step 5-Compute Shared Key

- Alice computes the shared key.

$$s = (B \bmod p)^a \bmod p$$

$$s = g^{b \cdot a} \bmod p$$

$$s = 3^{54 \cdot 24} \bmod 17$$

$$s = 1$$

- Bob computes the shared key.

$$s = (A \bmod p)^a \bmod p$$

$$s = g^{a \cdot b} \bmod p$$

$$s = 3^{24 \cdot 54} \bmod 17$$

$$s = 1$$

- Alice & Bob have created a shared secret key, **s**, unknown to Trudy



Diffie Hellman Algorithm

- Attacker Trudy only knows p , g , A and B , and so cannot compute S .
- For best result, $(p-1)/2$ should also be a prime and g should be a primitive root of p .
 - This means: $g^x \% p$, $1 < x < p$ takes all the values between 1 and p .
 - Thus, all possible values are used.
- This scheme is vulnerable to a man-in-the-middle attack.
 - Just like other public key systems.



Security of Diffie Hellman

- The Diffie-Hellman algorithm relies on the hardness of the discrete logarithm problem for its security.
- The discrete logarithm problem states that if we know p, g and public key $A = g^x \% p$, then it is computationally infeasible to calculate the private key a .
- It can be shown that this problem is equivalent to the factoring problem.
 - If a fast algorithm for factoring is discovered then there exists an equivalent fast algorithm for solving the discrete logarithm problem, and vice versa.



Summary

- A big picture of Public key Cryptography
 - A party owns a key pair (Public key, Private key)
 - Encrypt using Public key, decrypt using Private key

- Diffie Hellman Key Exchange protocol
 - Public channels can be used to create a confidential shared key function.
 - One-way operation should be used
 - Mix colour example
 - In practice: we use multiplicative group of integers modulo p , where p is prime, and g is a primitive root modulo p



Quizzes

1. Describe in detail the Diffie-Hellman method whereby Alice and Bob can agree on a common key.